`cover_descent` rows repriced at the GHS genus floor

2026-10-15. Generator: `src/cryptanalysis/jacobian_transfer/mod.rs (ghs_genus_floor)`. Output: `docs/curves/jacobian-transfers.json`, rebuilt by `jacobian_transfers build` and checked by `jacobian_transfers check`. Counts below come from `tabulate.py`, which reads the previous and the new committed file and counts. Source of the floor: `research/odd_char_descent_genus3_20261014/`.

Result (exact computation on the committed file)

Every `cover_descent` row on a `GF(p^k)` curve was priced at genus `k`. No GHS cover of `E/GF(p^k)` with `F_p(j) = GF(p^k)` has genus below 9, 49 and 41 at `k` = 6, 10, 14, so those rows are now priced at that floor. The table counts the rows by their class before and after.

k	genus	class at genus k	rows	now advantage	now marginal	now boundary
6	6 ⟶ 9	`formula_advantage`	598	576	22	0
6	6 ⟶ 9	`formula_marginal`	37	0	22	15
10	10 ⟶ 49	`formula_advantage`	126	0	0	126
10	10 ⟶ 49	`formula_marginal`	4	0	0	4
14	14 ⟶ 41	`formula_advantage`	122	23	19	80
14	14 ⟶ 41	`formula_marginal`	5	0	0	5
all		`formula_advantage`	**846**	**599**	**41**	**206**
all		`formula_marginal`	**46**	**0**	**22**	**24**

Curve verdicts (best class over a curve's routes), old file ⟶ new file:

verdict before	verdict after	curves
`boundary`	`boundary`	1201
`formula_advantage`	`boundary`	206
`formula_advantage`	`formula_advantage`	601
`formula_advantage`	`formula_marginal`	41
`formula_marginal`	`boundary`	24
`formula_marginal`	`formula_marginal`	26

`formula_advantage` verdicts go from 848 to 601. Rows the model still favours at the floor genus keep their class; the rest are demoted by the model's own pricing, not by a new rule.

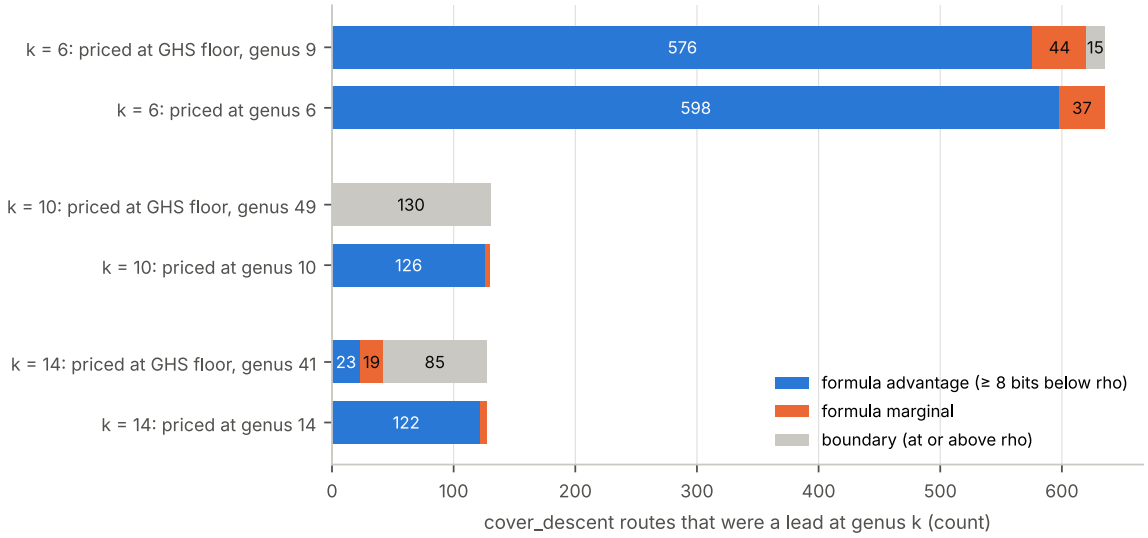


Figure 1: `cover_descent` rows that were a lead at genus `k`, by class before and after pricing at the GHS floor. Source: `route_changes.csv`.

Claims and scope

- **Exact computation:** the counts above, on the regenerated file, which `jacobian_transfers check` reproduces from `registry.json`.
- **From the source study:** the floor values (a proof for the lower bound, an exact enumeration for the table) and Joux–Vitse App. C for genus 9 at `k` = 6.
- **Not shown:** that any cover exists. The rows stay `proposed`. The floor is a lower bound for GHS covers only. Whether a particular curve attains it is verified only at `k` = 6; a curve whose `j` lies in a proper subfield can have a smaller cover; a non-GHS construction is not covered.

- **Not shown:** that the surviving advantages are real. The cost model prices a genus-`g` operation at `10 g²` multiplications, which counted textbook Cantor exceeds by 2 to 3.3 bits (`research/jacobian_op_count_20261014/`), so rows within a few bits of the margin could move.
- **Correction recorded:** an earlier reading was that the 846 advantage verdicts "rest on a genus no GHS cover has", which suggested they should all be demoted. Pricing at the genus the construction does have leaves 599 of the 846 rows as advantage; the demotions are the `k = 10` rows and most of the `k = 14` rows.